The correct option is **D**

$Cu\,(OH)_2.\,CuCO_3$

$$2KClO_3(s) \xrightarrow[\text{catalyst}]{\text{heat}} 2KCl(s) + 3O_2(g)$$

In the above reaction, potassium chlorate($KClO_3$) decomposes to form potassium chloride(KCl) and oxygen(O_2). Also as shown in the equation, heat is supplied for the reaction to take place.

Therefore, it is a decomposition reaction which is also endothermic in nature.

The correct option is **C**

$CaSO_4.2H_2O$

The correct option is **A** mercury
An ore consists of desired metal compound from which the metal can be extracted profitably.

"Carbonate ores get changed into oxides by heating strongly in limited air. This process is known as calcination."
Calcination is a process in which ore is heated in the absence of air.

Silver chloride to be stored in dark colored bottles:

- When exposed to sunshine, silver chloride disintegrates into silver and chlorine gas.
- Dark-colored bottles block the flow of light, preventing light from reaching silver chloride in the bottles and preventing its disintegration.
- In the presence of sunlight, silver chloride disintegrates to silver and chlorine, which function as a filter, preventing the reactions.
- As a result, silver chloride is kept in dark-colored vials.

Step 2: Oxidation of alcohol:

In the next step, the alcohol in presence of oxidizing agent gives aldehyde as an intermediate, and then finally carboxylic acid

The reaction is as follows,

$$CH_3CH_2OH\left(1\right)\overset{\text{Alk KMnO}_4}{\rightarrow} CH_3CHO\left(1\right)\overset{H^+}{\rightarrow}$$

$$CH_3COOH\left(1\right)$$

Step 1: Esterification:

As in step first, there is the reaction of a carboxylic acid $C_2H_4O_2$ and an alcohol in the presence of a few drops of sulphuric acid

The reaction is as follows,

$$CH_3COOH\left(1\right) + CH_3CH_2OH\left(1\right) \xrightarrow{H_2SO_4} CH_3COOCH_2CH_3\left(1\right)$$

Step 2: Oxidation of alcohol:

In the next step, the alcohol in presence of oxidizing agent gives aldehyde as an intermediate, and then finally carboxylic acid

The reaction is as follows,

$$CH_3CH_2OH\left(1\right) \overset{Alk\ KMnO_4}{\rightarrow} CH_3CHO\left(1\right) \overset{H^+}{\rightarrow} CH_3COOH\left(1\right)$$

On the basis of physical properties-

Ethanol	Ethanoic acid
The melting point of ethanol is 156 K.	The melting point of ethanoic acid is 289 K.
Ethanol has a pleasant odour	Ethanoic acid has a vinegar-like smell.

On the basis of chemical properties-

The compound which produces brisk effervescence, when reacting with aqueous solution of NaHCO3, must be ethanoic acid, whereas, the one which will show no effect will be ethanol.

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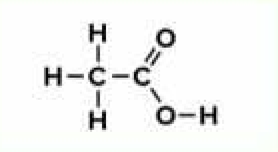
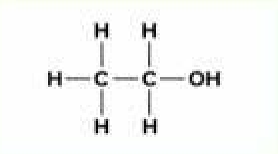
Following are the compounds reacted and formed,

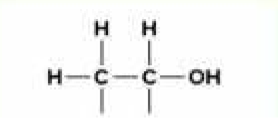
Carboxylic acid is **ethanoic acid (**$\mathrm{CH_3COOH}$**).**



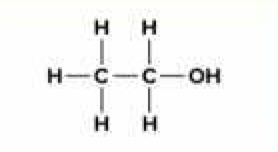
Alcohol is ethanol ($\mathrm{CH_3CH_2OH}$)



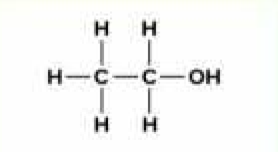
X is ethyl ethanoate ($\mathrm{CH_3COOCH_2CH_3}$)



Alcohol is ethanol (CH_3CH_2OH)



X is ethyl ethanoate ($CH_3COOCH_2CH_3$)



Hence ethyl ethanoate is formed by the reaction of ethanol and acetic acid (ethanoic acid).

A

red litmus turns blue.

Ash from magnesium ribbon is magnesium oxide. When this oxide is dissolved in water, it liberates hydroxyl [OH^-] ions in water. This makes the solution basic. Hence, we know in the basic solution turns red litmus into blue.

Sodium chloride is used for manufacture of sodium hydroxide which is called chloralkali process. In this process chlorine and hydrogen gas are formed as byproducts along with sodium hydroxide. Chlorine gas gives bleaching power when reacts with lime water and used as bleaching agent in chemical industries.

$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H$

$X \rightarrow Cl_2$ (Chlorine gas)

$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$

Therefore,

- The gas X is chlorine.
- Compound Y is calcium oxychloride which is commonly known as bleaching powder and used as bleaching agent in chemical